

Level	OBJECTIVES & REQUIREMENTS	Objective or Requirement	Date Last	Reasoning	Source(s)	System/	Status?	ASSOCIATED RISKS	Associated Risks with	VERIFICATION & VALIDATION (V&V)	Testing Method	Verification Plan	Validation	Any Pre-	Notes For
		Full statement sentence. Objectives are the requirements in simple, easy to understand, general terms. Requirements are technical at "shall" statements with a specific, measurable, and testable guideline to be met.	Date of when the requirement statement was last revised.		Where did this requirement come from? Is this based on literature, stakeholder, testing standards, etc? Include links if possible.	Does this matter on a system level? If not, which subsystems does this directly affect? Consider the interfaces as well. Which subsystems are responsible?	Accomplished or unaccomplished?		List the associated risk numbers.			Verification - Does it meet the specification need? Are we building our product right?	Validation - Does it meet the customer need? Did we build the right product?		What work did you do for this requirement this semester
OBJ = Objective, REQ = Requirement, Constraint, Number Hierarchy = 01.02.03 01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc) System, Subsystem, or Component Level?		The multi-agent Europa system shall operate fully autonomously for the entire surface mission duration without requiring real-time command uplinks from Earth.	6/14/2026	The 44-minute one-way light delay between Earth and Jupiter makes real-time demanding physically impossible, all decisions must be onboard. Traceable to OBJ.01	NASA FY 2026-2027 SBIR Solicitation, Topic COMNAV.1.S26B, 2025	System	Unaccomplished		RISK-01: Single-point autonomy failure with no fallback human override		Hardware-in-the-loop simulation. Run a 72-hour autonomous mission sequence with a simulated 44-min communication blackout, and verify zero unresolved command dependencies on Earth.	Verify the software autonomously executes all planned traverses, consensus rounds, and science data collected without any Earth uplink during the blackout window.	Demonstrate that the mission objectives are achieved across three independent HILL runs without Earth support		
REQ.01.02		The system shall survive and maintain operational status after exposure to total ionizing dose of no less than 100 krad over the nominal mission lifetime.	6/14/2026	Jupiter's magnetosphere delivers radiation doses far exceeding near-Earth missions. Europas Clipper's radiation vault architecture targets ~100 krad as the design threshold. Radiation environment directly drives the shielded vault requirement. Traceable to OBJ.01	National Academies, Origins Worlds and Life Decadal Survey 2023–2032, https://www.nationalacademies.org/read/26522-chapter/10 ; Galileo spacecraft radiation vault design heritage	System, Electronics, Power	Unaccomplished		RISK-02: Accumulated single event upsets degrade consensus algorithm convergence before mission completion.		Co-60 gamma irradiation of all radiation-sensitive components to 110 krad (10% margin). Monitor functional performance throughout exposure.	Verify all components pass functional tests at end of life dose with no loss of required functionality	Validate the integrated rover and relay nodes complete simulated mission operations after T1D exposure.		
REQ.01.03		The system shall maintain a minimum inter-agent network connectivity of two active relay nodes at all times during surface operations to ensure consensus is achievable.	6/14/2026	Consensus algorithms require a minimum connected graph to guarantee convergence. Two active relays plus a surface rover guarantee a 3-node connected topology as the minimum viable configuration. Traceable to OBJ.01 and OBJ.04.	NASA STMD Civil Space Technology Shortfall Rankings, Jul. 2024, https://www.nasa.gov/wp-content/uploads/2024/07/civil-space-shortfall-ranking-july-2024.pdf	System, COMMs	Unaccomplished		RISK-03: Simultaneous radiation-induced failure of multiple relay nodes isolating the surface rover.		Network fault simulation: simulate progressive relay node failures and verify consensus detects degenerated topology and triggers reconfiguration before dropping below 2 active nodes.	Verify that fault detection and reconfiguration logic activates within one consensus cycle period upon detection of a node loss event.	Validate that the surface rover completes a traverse plan using map data from a degraded 2-relay configuration.		
REQ.02.01		The surface robot shall maintain a localization accuracy of no worse than +0.5 m relative to its last known position during any traverse segments up to 50 m in the absence of GPS.	6/14/2026	Cryobot insertion site selection requires meter-level positioning to match the radar identified thin ice candidate. +0.5 over 50 m segments is consistent with visual odometry performance on Mars rovers. Traceable to OBJ.02	NASA FY 2026-2027 SBIR Solicitation, Topic COMNAV.1.S26B, Mars Curiosity rover visual odometry heritage	Mobility, GNC	Unaccomplished		RISK-04: Ice surface texture homogeneity reducing visual odometry feature tracking quality.		Field test on analog terrain: command 50m traverse segments and compare onboard state estimate to ground truth GPSD position across 20 runs under varying terrain conditions.	verify that position error remains within +0.5 m across a minimum of 20 traverse segments.	Validate that the rover navigates to and stops within +0.5 m of a designated waypoint representing a candidate insertion site.		
REQ.03.01		The ground-penetrating radar sounder shall resolve ice shell thickness to within +10% accuracy for ice layers between 1 km and 30 km depth, producing a subsurface map at the minimum spatial resolution of 100 m along the traverse track.	6/14/2026	Identifying optimal cryobot insertion site requires knowing where the ice shell is the thinnest. +- 10% thickness accuracy is enough to distinguish viable drilling candidates. The 100 m along track resolution is consistent with the Europa Clipper REASON instrument design parameters. Traceable to OBJ.03	National Academies, Origins Worlds and Life Decadal Survey 2023–2032, https://www.nationalacademies.org/read/26522-chapter/10 ; Europa Clipper REASON instrument documentation	Science/Payload, Power	Unaccomplished		RISK-05: Ice shell heterogeneity or brine inclusions creating radar clutter that degrades thickness estimates.		Laboratory radar range measurements against ice targets of known thickness using dielectric matches analogs, verify +10% accuracy across the 1-30 km depth range.	Verify that the radar sounder produces subsurface profiles with +10% depth accuracy against calibration targets before rover integration.	Validate that the subsurface map produces during a field traverse correctly identifies preserved thin ice candidate sites within the required spatial resolution.		
REQ.04.01		The inter-agent mesh network shall achieve consensus on a shared consensus on a shared terrain map update within no more than 10 minutes under a degraded network condition defined as packet loss rates of up to 30% on any single inter-agent link	6/14/2026	Rover traverse planning depends on a current consensus validated terrain map. A 10 minute convergence bound limits traverse planning delay to an acceptable operations cadence. 30% packet loss represents a plausible radiation induced degradation scenario for the environment. Traceable to OBJ.04.	NASA STMD Civil Space Technology Shortfall Rankings, Jul. 2024, NASA FY 2026-2027 SBIR Solicitation, Topic COMNAV.1.S26B	COMMs, CDH	Unaccomplished		RISK-06: Radiation induced burst errors exceeding 30% packet loss on multiple links simultaneously, stalling consensus.		Software-in-the-loop consensus simulation: Introduce 30% packet loss on each inter-agent link and measure convergence time across 100 test runs.	Verify the consensus convergence time remains <10 minutes across all 100 test runs with 30% packet loss.	Validate that the rover initiates a traverse using a map update produced by the consensus protocol under the degraded link test condition.		
REQ.03.02		The radar sounder antenna shall be mounted such that the antenna boresight remains within +5 of nadir during all active sounding operations; the rover shall halt traversal if the measured surface slope exceeds 20 to maintain pointing compliance.	6/16/2026	A nadir pointing antenna is required to minimize geometric error in the two-way travel time measurement. Off nadir tilt reduces basal reflection return strength and introduces range error that degrades	National Academies, Origins Worlds and Life Decadal Survey 2023–2032; Europa Clipper REASON instrument documentation	Science/Payload, Structures, Mobility	Unaccomplished		RISK-05: Ice shell heterogeneity creating radar clutter that degrades thickness estimates; RISK-04: Ice surface texture reducing visual odometry quality (slope-stop logic dependency)		Bench test antenna mounting fixture: tilt rover chassis to 5° increments and measure antenna boresight deviation using an inclinometer. Verify pointing error stays within ±5° across the full range of surface slopes up to 20°.	Verify that antenna boresight deviation does not exceed ±5° of nadir across all chassis tilt angles up to 20° slope	Validate that the rover correctly halts traversal and resumes sounding when slope threshold is exceeded during an analog terrain field test.		
REQ.03.03		The rover power subsystem shall reserve a dedicated peak power allocation for radar sounder transmission pulses and shall command drive motor shutdown during each active sounding window to prevent power contention with the RTG continuous output.	6/16/2026	Radar sounders draw high peak pulse power during transmission. The RTG provides fixed continuous output; simultaneous drive and sounding would exceed available power margin and risk under-powering the radar pulse, degrading return signal amplitude below the noise floor. Stop-and-sound is consistent with high-power instrument operation heritage on Mars rovers. Traceable to OBJ.03. At approximately 40 traces per sol across four data sheets, raw science data accumulates faster than the non-deterministic Earth relay can downlink. Seven sols of buffer covers the consensus convergence window and plausible communication gaps. Lossless compression preserves the measurement fidelity required by REQ.03.01 while reducing relay load. Traceable to OBJ.03 and OBJ.04.	Europa Clipper REASON instrument documentation; NASA RTG power system heritage	Science/Payload, Power, Mobility	Unaccomplished		RISK-02: Accumulated single event upsets degrade system performance; RISK-05: Degraded radar return amplitude from insufficient transmission power		Hardware-in-the-loop power budget simulation: command simultaneous drive and sounding operations and monitor RTG output margin across 50 test cycles. Verify drive shutdown command fires before each sounding pulse.	Verify that drive motors are commanded off prior to every sounding transmission and that RTG output margin remains positive across all test cycles.	Validate that the rover completes a full simulated traverse sol with correct stop-and-sound behavior and no power fault events.		
REQ.03.04		The command and data handling subsystem shall provide onboard storage sufficient to retain a minimum of 7 sols of uncompressed radar trace data, and shall apply lossless compression to all Radar Traces records prior to inter-agent relay transmission, achieving a minimum compression ratio of 2:1.	6/16/2026	Europa's surface temperature is approximately 90K (-185°). Battery cells experience severe capacity loss below -20C and permanent damage below -40C. The RTG provides continuous waste heat that can be routed to a thermal enclosure, making active heating feasible without a dedicated power draw penalty. No ground intervention is possible to recover a thermally failed rover. Traceable to OBJ.01.	NASA STMD Civil Space Technology Shortfall Rankings, Jul. 2024; Europa Clipper REASON instrument documentation	Science/Payload, COMMs, CDH	Unaccomplished		RISK-06: Radiation induced burst errors exceeding 30% packet loss stalling consensus; RISK-05: Ice shell heterogeneity generating excess CLUTTER traces inflating data volume		Software-in-the-loop storage simulation: generate 7 sols of synthetic radar trace data at full volume and verify onboard storage does not overflow. Apply compression algorithm and measure output ratio across 100 data sets.	Verify that onboard storage accommodates 7 sols of uncompressed data and that compression ratio meets or exceeds 2:1 across all 100 test data sets.	Validate that compressed science data downlinked through the inter-agent mesh under 30% packet loss conditions can be fully reconstructed without data loss on the receiving node.		
REQ.05.01		The rover thermal control system shall maintain all avionics and battery cells within an operational temperature range of 0C to 40C during all active surface operations and above -20C during night-mode low-power thermal management, using active heater elements powered by RTG waste heat.	6/28/2026	Europa's surface temperature is approximately 90K (-185°). Battery cells experience severe capacity loss below -20C and permanent damage below -40C. The RTG provides continuous waste heat that can be routed to a thermal enclosure, making active heating feasible without a dedicated power draw penalty. No ground intervention is possible to recover a thermally failed rover. Traceable to OBJ.01.	National Academies, Origins Worlds and Life Decadal Survey 2023–2032; NASA RTG thermal heritage	Thermal, Power, Structures	Unaccomplished		RISK-07: Thermal control failure during extended night-mode causing battery cell damage and mission termination		Thermal-Vacuum chamber test: Subject the integrated rover to Europa analog thermal cycling (90K surface, simulated RTG waste heat input) and monitor avionics and battery temperatures across a full 23 hour sol cycle including night-mode.	Verify that avionics and battery temperatures remain within required bounds across three consecutive sol cycles in the thermal vacuum chamber with no active ground commands.	Validate that the rover completes a simulated surface traverse sol following a full night-mode thermal cycle without any subsystem degradation or heater fault events.		
REQ.05.02		The command and data handling subsystem shall detect and autonomously recover from a single-event latchup in any processor core within 330 seconds of occurrence, restoring nominal operations without ground intervention, and shall log each latchup event to non-volatile memory with a timestamp and affected core identifier.	6/28/2026	Jupiter's magnetosphere produces a high-energy particle flux that induces single event latchup at a far higher rate than near-Earth missions. With a 44-minute one-way light delay no ground intervention can recover a locked core in time to prevent missions tall. The 30-second recovery bound prevents autonomous traverse planning and site selection from stalling indefinitely. Traceable to OBJ.01, linked to RISK-02.	National Academies, Origins Worlds and Life Decadal Survey 2023–2032; NASA STMD Civil Space Technology Shortfall Rankings, Jul. 2024; Galileo spacecraft radiation heritage	CDH, Electronics	Unaccomplished		RISK-02: Accumulated single event upsets degrade consensus algorithm convergence before mission completion.		Hardware-in-the-loop fault injection: Inject simulated single-event latchup signals into each processor core and measure time from latchup detection to nominal operation restoration across 50 test runs per core.	Verify that the recovery from the latchup injection occurs within 30 seconds across all 50 test runs for each core, and that each event is correctly logged to non-volatile memory with timestamp and code ID.	Validate that the rover autonomously continues a planned traverse sequence without human intervention following a latchup event injected mid-traverse during an analog mission simulation.		
REQ.05.03		The rover power management subsystem shall enforce a per-sol energy budget such that total energy draw across all subsystems does not exceed 90% of the RTG's nominal daily energy output, reserving a minimum 10% margin for autonomous anomaly response, heater activations, and unplanned communication relay passes.	6/28/2026	The RTG provides fixed continuous output with no ability to surge on demand. Radiation-induced matchups, slope-stop halts per REQ.03.02, and degraded consensus requiring extra relay passes all introduce unscheduled power draws. A hard 10% reserve prevents the rover from entering a power fault state during autonomous recovery events with no ground support available. Traceable to OBJ.01 and OBJ.03	NASA RTG power system heritage; Europa Clipper power budget architecture; NASA FY 2026–2027 SBIR Solicitation, Topic COMNAV.1.S26B	Power, CDH, Thermal, Mobility	Unaccomplished		RISK-02: Accumulated single event upsets degrade consensus algorithm convergence before mission completion. RISK-07: Thermal control failure during extended night-mode causing battery cell damage and mission termination		Hardware-in-the-Loop power budget simulation: Run a full synthetic sol energy draw does not exceed 90% of RTG nominal output across all 50 test cycles, including cycles with injected anomaly events that trigger unscheduled heater and relay draws.	Verify that the total per-sol energy draw does not exceed 90% of RTG nominal output across all 50 test cycles, including cycles with injected anomaly events that trigger unscheduled heater and relay draws.	Validate that the rover completes a full simulated traverse sol with one injected latch up and one slope-stop event without triggering a power fault or dropping below that 10% reserve margin.		

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		OBJ = Objective, REQ = Requirement, Constraint, CSR = Requirement, Number Hierarchy = 01.02.03 (01 = Objective Parent Category, 02 = First Child Requirement, 03 = Second Child Requirement... etc)	System, Subsystem, or Component Level?	Full statement sentence. Objectives are the requirements in simple, easy to understand, general terms. Requirements are technical as "shall" statements with a specific, measurable, and testable guideline to be met. Date of when the requirement statement was last revised.	Where did this requirement come from? Is this based on literature, stakeholder, testing standards, etc? Include links if possible.	Does this matter on a system level? If not, which subsystems does this directly affect? Consider the interfaces as well. Which subsystems are responsible?	Accomplished or unaccomplished?				List the associated risk numbers.	Verification - Does it meet the specification need? Are we building our product right?	Validation - Does it meet the customer need? Did we build the right product?	What work did you do for this requirement this semester			
REQ.06.01		Subsystem [Operation Level] Mobility	The rover shall autonomously plan and execute a surface traverse of no less than 100m per sol across Europa analog terrain, pausing at intervals no greater than 100m along-track to perform a radar sounding stop, and shall resume traversal upon sounding completion without requiring a ground uplink.	6/28/2026	The mission objective requires building a spatially continuous subsurface ice map at <100m along-track resolution (REQ.03.01). To meet that resolution the rover must stop and sound at least every 100m, then continue. With no real-time Earth support (REQ.01.01), the stop-and-sound cycle must be fully onboard. The 100m/sol minimum traverse rate ensures the science map grows at a pace consistent with the 30-sol surface operations window. Traceable to OBJ.02 and OBJ.03.	National Academies, Origins Worlds and Life Decadal Survey 2023–2032; Europa Clipper REASON instrument documentation; NASA FY 2026–2027 SBIR Solicitation, Topic COMNAV.1.S26B	Mobility, GNC, Science/Payload				Unaccomplished	RISK-05: Ice shell heterogeneously creating radar clutter that degrades thickness estimates; RISK-04: Ice surface texture reducing visual odometry quality (slope-stop logic dependency)	Field test on analog terrain: command the rover to execute a 100m traverse with stop-and-sound events at <100m intervals and measure total traverse distance completed and number of sounding stops executed across 10 runs.	Verify that the rover completes >100m of traverse per simulated sol and executes a sounding stop at every <100m interval across all 10 test runs without a ground uplink command.	Validate that the subsurface map generated from a full simulated traverse sol contains valid radar traces at <100m along-track spacing covering the full traverse distance.		
REQ.06.02		Subsystem [Operations Level] Science / Payload	The rover's onboard site-selection algorithm shall autonomously evaluate each NOMINAL-flagged radar trace against the cryobot insertion criteria (ice thickness < 5 km, surface slope < 20, terrain roughness index <0.7) and shall flag a candidate insertion site within one sol of the criteria first being satisfied without a ground intervention.	6/28/2026	The terminal mission objective is identifying an optimal cryobot insertion site autonomously. With a 44-minute light delay, Earth cannot evaluate radar data and issue a site selection decision in real time. The onboard algorithm must continuously evaluate incoming traces and trigger the site flags as soon as conditions are met. The one-sol response bound ensures the mission does not waste traverse budget driving past a viable site. Traceable to OBJ.03.	National Academies, Origins Worlds and Life Decadal Survey 2023–2032; Europa Clipper REASON instrument documentation; NASA FY 2026–2027 SBIR Solicitation, Topic COMNAV.1.S26B	Science/Payload, CDH				Unaccomplished	RISK-05: Ice shell heterogeneously creating radar clutter that degrades thickness estimates;	Software-in-the-loop simulation: inject synthetic radar trace datasets containing pre-seed candidate sites at known positions and measure whether the onboard algorithm flags the correct site within one sol of the criteria being met across 50 test runs.	Verify that the site-selection algorithm correctly flags all pre-seeded candidate sites within one sol of criteria satisfaction and produces zero false positive flags on CLUTTER or NOISE_FLOOR traces across all 50 runs.	Validate that during an analog terrain field traverse, the algorithm flags a candidate site matching pre-surveyed thin-ice analog targets within one sol of the rover reaching the qualifying zone.		
REQ.06.03		Subsystem [Operations Level] COMMS	The rover shall autonomously transmit a daily science summary packet to the orbital relay network containing the sol's Daily Summary record (traverse distance, valid trace count, minimum ice thickness, candidate site count, and mean consensus convergence time) within 2 hours of completing each sol's radar data collection window, without requiring a ground-initiated downlink command/	6/28/2026	Earth-based scientists and mission planners require timely sol-by-sol progress data to assess whether mission objectives are being met and to prepare contingency plans within the non-deterministic relay window. With a 44-minute light delay, the rover cannot wait for a downlink command; it must autonomously initiate transmission when the data is ready. The 2-hour bound ensures the daily summary reaches before the next sol's traverse planning begins, maintaining operational cadence. Traceable to OBJ.01 and OBJ.04	NASA STMD Civil Space Technology Shortfall Rankings, Jul. 2024; NASA FY 2026–2027 SBIR Solicitation, Topic COMNAV.1.S26B	COMMS, CDH, Science/Payload				Unaccomplished	RISK-06: Radiation induced burst errors exceeding 30% packet loss stalling consensus; RISK-05: Ice shell heterogeneously creating radar clutter that degrades thickness estimates	Earth-based scientists and mission planners require timely sol-by-sol progress data to assess whether mission objectives are being met and to prepare contingency plans within the non-deterministic relay window. With a 44-minute light delay, the rover cannot wait for a downlink command; it must autonomously initiate transmission when the data is ready. The 2-hour bound ensures the daily summary reaches before the next sol's traverse planning begins, maintaining operational cadence. Traceable to OBJ.01 and OBJ.04	Verify that the daily summary packet is autonomously transmitted and confirmed received at the relay node within 2 hours of data collection completion across all 30 test runs, including degraded-link runs.	Validate that a complete daily summary record is successfully reconstructed from a compressed packet received through the inter-agent mesh under 30% packet loss conditions during an end-to-end mission simulation.		